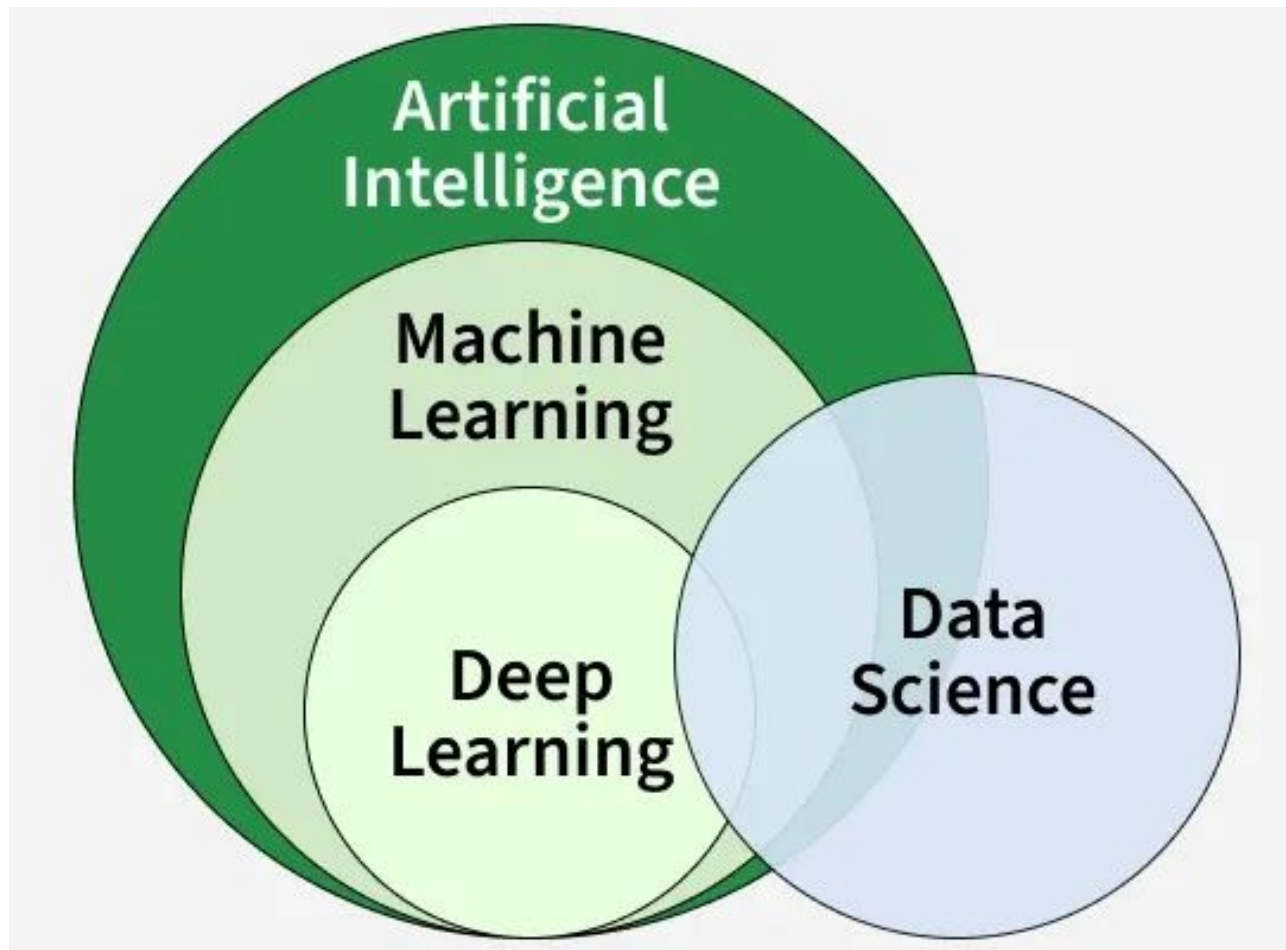


Deep Learning



Machine Learning



Uses algorithms and learns on its own but may need human intervention to correct errors

Deep Learning Process



Understands problem and whether deep learning is a good fit



Trains algorithm on large amount of labeled cats



Identifies relevant data sets and prepares them for analysis

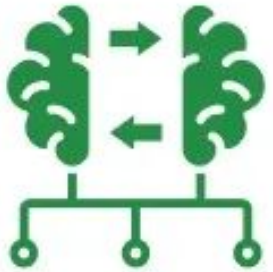


Tests the model's performance against unlabeled data



Chooses the type of deep learning algorithm to use

TYPES OF NEURAL NETWORKS



Feedforward
neural networks
(FNNs)



Convolutional
Neural Networks
(CNNs)



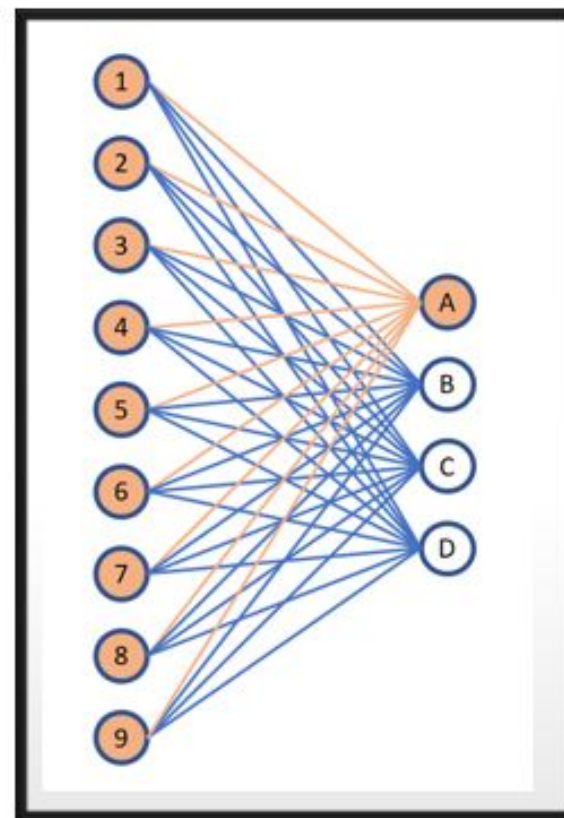
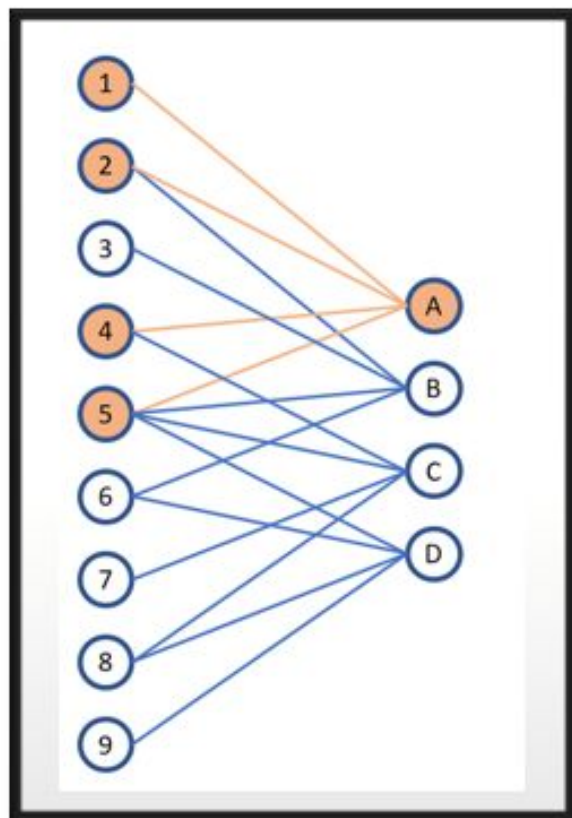
Recurrent Neural
Networks (RNNs)



Generative
Adversarial
Networks (GANs)



Autoencoders,
Transformer
Networks



Fully vs Partially Connected Layers

Fully Connected Layers (Dense Layers)

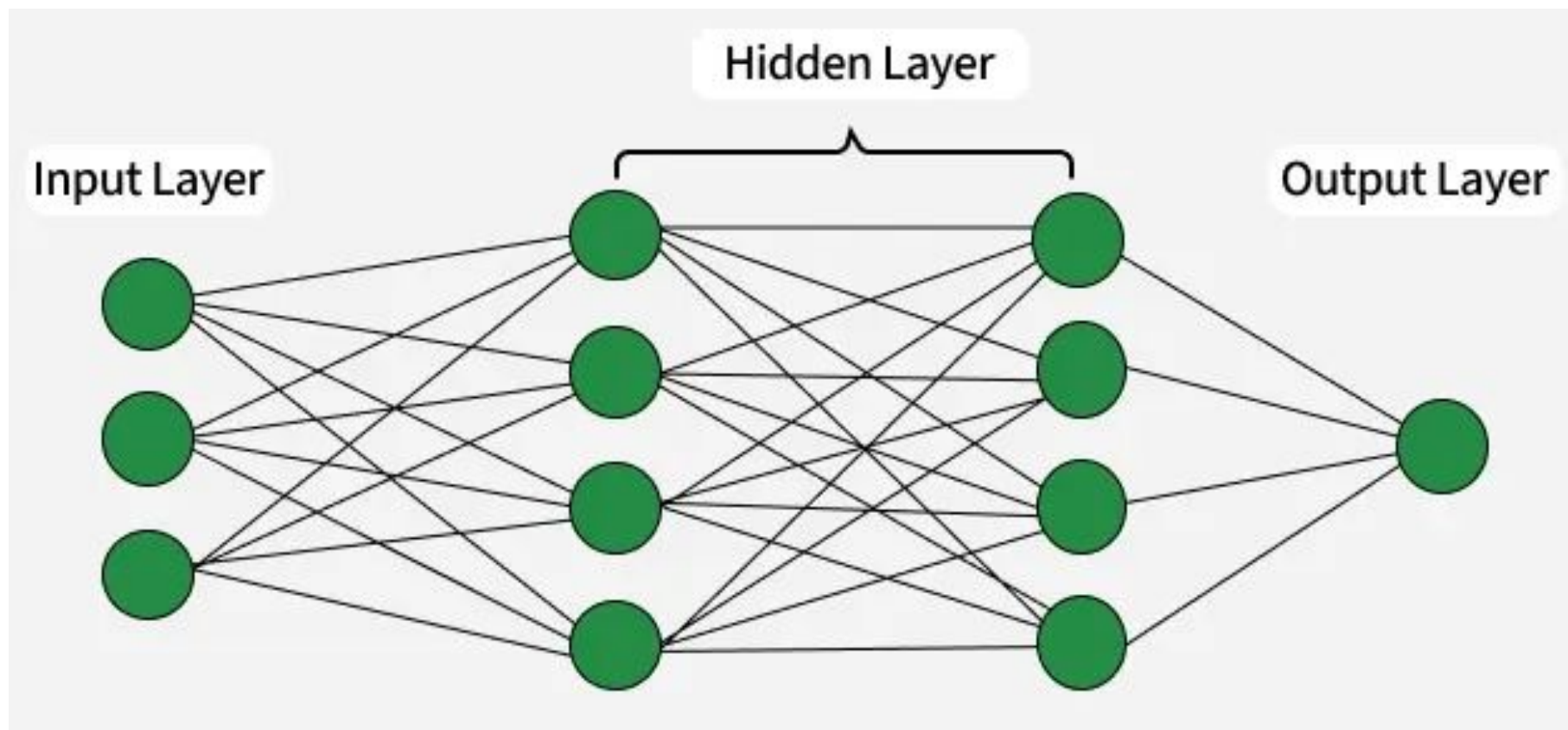
- ▶ **Connectivity:** Each neuron connects to every neuron in the preceding layer.
- ▶ **Parameters:** High number of weights, computationally expensive.
- ▶ **Function:** Learns global patterns and complex relationships, often used in the final classification stages.
- ▶ **Example:** General Purpose Neural Networks.

Partially Connected Layers

- ▶ **Connectivity:** Each neuron connects only to a local region (receptive field) of the input, not the entire input.
- ▶ **Parameters:** Fewer parameters due to weight sharing (the same filter/kernel is applied across the input).
- ▶ **Function:** Efficiently extracts spatial features (edges, textures) from data like images.
- ▶ **Example:** Convolutional Neural Networks (CNNs) use these for feature extraction.

Deep learning

- ▶ Deep Learning is transforming the way machines understand, learn and interact with complex data. Deep learning mimics neural networks of the human brain, it enables computers to autonomously uncover patterns and make informed decisions from vast amounts of unstructured data.
- ▶ Neural network consists of layers of interconnected nodes or neurons that collaborate to process input data.
- ▶ In a fully connected deep neural network data flows through multiple layers where each neuron performs nonlinear transformations, allowing the model to learn intricate representations of the data.
- ▶ The input layer receives data which passes through hidden layers that transform the data using nonlinear functions.
- ▶ The final output layer generates the model's prediction.



Applications

- ▶ Data Science
- ▶ Cyber Security
- ▶ Autonomous Vehicles
- ▶ Social Media Analytics
- ▶ Healthcare and Medical Sector
- ▶ Computer Vision
- ▶ NLP, Speech and Audio

Aspect	Machine Learning	Deep Learning
Basic Idea	Applies statistical algorithms to learn patterns from data	Uses artificial neural networks to learn patterns from data
Data Requirement	Works well with small to medium datasets	Requires a large amount of data
Task Complexity	Better for simple and low-label tasks	Better for complex tasks like image and text processing
Training Time	Takes less time to train	Takes more time to train
Feature Extraction	Features are manually selected and extracted	Features are automatically extracted
Learning Process	Not end-to-end	End-to-end learning
Model Complexity	Less complex	Highly complex
Interpretability	Easy to understand and explain	Hard to interpret (black box)
Hardware Requirement	Can run on CPU, needs less computing power	Needs GPU and high-performance systems
Use Cases	Spam detection, recommendation systems	Image recognition, NLP, speech recognition